

Yoshini Bailung

Curriculum Vitae

PERSONAL DETAILS

DESIGNATION: PhD, Post-Doctoral Fellow
ADDRESS: 4 Rue Enrico Fermi, Paul Dirac Building,
Claude Bernard University Lyon 1, 69622, Villeurbanne
NATIONALITY: Indian
EMAIL: yoshini.bailung@cern.ch; y.bailung@ip2i.in2p3.fr
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PROFESSIONAL EXPERIENCE

2025-present CNRS Post-Doctoral Researcher at the
Institut de Physique des Deux Infinis (IP2I) de Lyon

2024-2025 Research Associate at Indian Institute of Technology Indore

EDUCATION

2019-2025 Ph.D. in Physics at Indian Institute of Technology Indore
THESIS TITLE: “Charm production in proton–proton collisions with ALICE at the LHC and light nuclei synthesis via coalescence in heavy-ion collisions”
[Link to contribution](#)
Date of defense: 11th February 2025
SUPERVISOR: Prof. Ankhi Roy

2017-2019 Master of Science (M.Sc.) in Physics from Gauhati University
THESIS TITLE: “Cabibbo motivated 3+1 Neutrino Mixing”
SUPERVISOR: Dr. Subhankar Roy

2014-2017 Bachelor of Science (B.Sc.) in Physics from SGTB Khalsa College,
University of Delhi

RESEARCH ACTIVITIES

1. Experimental Data Analysis with ALICE@CERN:

(a) Beauty-jet measurement - IP2I/CNRS, 2025 - Present

Developed a machine-learning-based b-jet tagger for charged-jets in pp collisions at $\sqrt{s} = 13.6$ TeV using LHC Run-3 data. Boosted Decision Trees were trained and validated to identify b-jets based on constituent track impact parameter information. Developed 2-dimensional discriminators (heavy flavour (bc) vs light flavour (udsg) and beauty (b) vs charm (c)) and performed 2D template fits for flavour based yield extraction. Production cross-section of b-jets was obtained after correcting and unfolding the measured spectrum to the particle level, and estimating the associated systematics. Measurement compared to the traditional secondary vertex method and a graph-based neural network (GNN) tagger. Monte Carlo comparisons were performed with PYTHIA8 + POWHEG calculations. Results were approved and presented at multiple international conferences.

(b) **Charm production measurement - IITI, Ph.D., 2019-2025:** Focused on measuring heavy flavour hadron production in pp collisions at $\sqrt{s}=13$ TeV using LHC Run-2 data. Charm yields were extracted via D-meson reconstruction from their hadronic decay channels. The analyses targeted D^0 , D^+ , and D^{*+} meson production, studying their production as a function of charged-particle multiplicity and transverse sphericity in pp collisions at $\sqrt{s}=13$ TeV. Focused on the optimization of the reconstruction and selection of the D^0 meson, the evaluation of the corrections for the measurement and the estimation of the relevant systematic uncertainties. Contributed to the Monte Carlo comparison study on the final results with PYTHIA8. The results of these studies were approved as preliminary results and presented at multiple international conferences.

2. **Phenomenology of Heavy-Ion Collisions - IITI, Ph.D., 2019-2025:**

Led studies on light-nuclei production using the coalescence mechanism. Developed a coalescence afterburner, initially using a simplified box approach, later refined with quantum mechanical wave functions. Contributed to multiple papers on heavy flavour phenomenology. First author of three papers on the coalescence as a primary mechanism of light-nuclei formation and co-author of three heavy flavour phenomenology papers. Short description of selected publications:

- “Exploring light nuclei production at RHIC and LHC energies with A Multi-Phase Transport model and a coalescence afterburner” [Nucl. Phys. A **1037**, 2023] – implemented a box coalescence model to explain light-nuclei yields in heavy-ion collisions.
- “Searching for enhancement in coalescence of in-jet (anti-)deuterons in proton-proton collisions” [Phys. Rev. C **109**, 2024] – developed a Wigner coalescence model, integrated into PYTHIA8 along with a nucleon emission source model, focusing on in-jet deuteron production.
- “Probing coalescence of light nuclei via femtoscopy and azimuthal anisotropies” [Phys. Rev. C **111** (2025) 014911] – used a correlation afterburner (CRAB) to study momentum correlations between nuclei clusters, estimating emission sources, formation times and anisotropic flow coefficients.

3. **Other activities:**

- Participation in the Run 3 data-taking activities as shifter for the ALICE Experimental Control System (ECS), on-call detector expert for the Muon Forward Tracker (MFT), and Shift Leader in Matter of Safety (SLIMOS).
- Involved in ALICE service work with the Zero Degree Calorimeter (ZDC) team - developed code to predict and reconstruct truncated ZDC signals using Machine Learning (ML) algorithms.
- Contributed to ALICE Data Processing Group’s (DPG) asynchronous quality control (QC) of Global Tracks.
- Participation in the institutional reviewer committee for three ALICE publications.
- Mentor/Co-supervisor of 1 Ph.D student, 5 Master’s students, 1 Bachelor’s project with thesis/reports on different topics as described below.

- Active in outreach activities related to scientific communication.

PUBLICATIONS

Co-author of 209 publications [InspireHEP](#)

h-index: 37

Citations: 4,580

Selected summary of publications where I am a primary contributor.

1. **Yoshini Bailung**, Sudhir P. Rode, Neha Shah, Ankhi Roy; “*Probing coalescence of light nuclei via femtoscopy and azimuthal anisotropies*”; Published in **Physical Review C**; [[doi:10.1103/PhysRevC.111.014911](#)]
2. **Yoshini Bailung**, Neha Shah, Ankhi Roy; “*Searching for enhancement in coalescence of in-jet (anti-)deuterons in proton–proton collisions*”; Published in **Physical Review C**; [[doi:10.1103/PhysRevC.109.044908](#)]
3. **Yoshini Bailung**, Neha Shah, Ankhi Roy; “*Exploring light nuclei production at RHIC and LHC energies with A Multi-Phase Transport model and a coalescence afterburner*”; Published in **Nuclear Physics A**; [[10.1016/j.nuclphysa.2023.122701](#)]
4. S. Khade, **Yoshini Bailung**, R. Gupta, R. Singh, A. Roy, “*Forward azimuthal correlations in pp and p-Pb collisions*”, Published in **European Physics Journal A**, 10.1140/epja/s10050-026-01924-2
5. S. Khade, R. Singh, **Yoshini Bailung**, A. Roy, “*An impact of parton distribution functions on open heavy flavour azimuthal angular correlations in Pb-Pb collisions at $\sqrt{s_{NN}} = 5.5$ TeV*”, Published in **Nuclear Physics A**, 10.1016/j.nuclphysa.2025.123220
6. Ravindra Singh, **Yoshini Bailung**, Sumit Kumar Kundu, Ankhi Roy; “*Jet fragmentation via azimuthal angular correlations of electrons from heavy flavor decay in pp, p-Pb, and Pb-Pb collisions using PYTHIA8+Angantyr calculations*”; Published in **Physical Review C**; [[10.1103/PhysRevC.107.024911](#)]
7. Sumit Kumar Kundu, **Yoshini Bailung**, Sudhir Pandurang Rode, Partha Pratim Bhaduri, Ankhi Roy; “*Effect of various particlization scenarios on anisotropic flow and particle production using UrQMD hybrid model*”; Published in **Nuclear Physics A**; [[10.1016/j.nuclphysa.2022.122574](#)]
8. Ravindra Singh, **Yoshini Bailung**, Ankhi Roy; “*Dynamics of particle production in Pb–Pb collisions at $\sqrt{s_{NN}} = 2.76$ TeV using PYTHIA8 Angantyr model*”; Published in **Physical Review C**; [[10.1103/PhysRevC.105.035202](#)]
9. Sumit Kumar Kundu, **Yoshini Bailung**, Sudhir Pandurang Rode, Partha Pratim Bhaduri, Ankhi Roy; “*Dependence on beam energy and nuclear equation of state of anisotropic flow and particle production in low-energy heavy-ion collisions*”; Published in **Physical Review C**; [[10.1103/PhysRevC.104.024907](#)]

ALICE ANALYSIS NOTES

1. **Yoshini Bailung**, Clement Lotteau, Cvetan Cheshkov ; “*Measurement of b-jet cross-section using Boosted Decision Trees in proton–proton collisions at $\sqrt{s} = 13.6$ TeV*”; ALICE Analysis Note; ID number: 1786; [[Link to contribution](#)]
2. Marco Giacalone, **Yoshini Bailung**, Randhir Singh ; “*Measurement of D-meson (D^0 , D^+ , D^{*+}) production as a function of event shapes in proton–proton collisions at $\sqrt{s} = 13$ TeV*”; ALICE Analysis Note; ID number: 1240; [[Link to contribution](#)]

3. Marco Giacalone, **Yoshini Bailung**, Randhir Singh ; “*Measurement of D-meson (D^0, D^+, D^{*+}) production versus charged particle multiplicity in proton-proton collisions at $\sqrt{s} = 13$ TeV*”; ALICE Analysis Note; ID number: ANA-1167; [[Link to contribution](#)]

CONFERENCE PROCEEDINGS

1. Rohit Kaundal, **Yoshini Bailung**, Swapnesh Khade, Ankhi Roy; “*A new look into ep collisions using event shape variables*” Published in **Proceedings of 68th DAE Symposium on Nuclear Physics, 7 - 11 December 2023** [[proceedings/snp2024](#)]
2. Swapnesh Khade, **Yoshini Bailung**, Rashi Gupta, Ravindra Singh, Ankhi Roy; “*Forward Physics of D^0 meson: A study of production and angular correlations in pp and p-Pb collisions*” Published in **Proceedings of 68th DAE Symposium on Nuclear Physics, 7 - 11 December 2023** [[proceedings/snp2024](#)]
3. **Yoshini Bailung**, Sudhir Pandurang Rode, Neha Shah, Ankhi Roy; “*Production of light nuclei in heavy-ion collisions via a coalescence mechanism*”; Published in **Proceedings of 67th DAE Symposium on Nuclear Physics, 9 - 13 December 2023** [[proceedings/snp2023](#)];
4. **Yoshini Bailung**, on behalf of ALICE Collaboration; “*Heavy-flavour production as a function of charged-particle multiplicity with ALICE at the LHC*”; Submitted to **MDPI Proceedings Series for Hot Quarks 2022, 11 – 17 October 2022** [Under review]
5. Sumit Kumar Kundu, **Yoshini Bailung**, Sudhir Pandurang Rode, Partha Pratim Bhaduri, Ankhi Roy; “*Dependence of anisotropic flow and particle production on particlization models and nuclear equation of state*”; Published in **Dynamics of HOT QCD Matter – Current status and Developments, International Journal of Modern Physics E, 12-14 May 2022** [[proceedings/HotQCD2022](#)]
6. Sumit Kumar Kundu, **Yoshini Bailung**, Sudhir Pandurang Rode, Partha Pratim Bhaduri, Ankhi Roy; “*Dependence of anisotropic flow of net-protons on particlization model for various nuclear equation of state*”; Published in **Proceedings of 65th DAE Nuclear Physics Symposium, 1-5 December 2021** [[proceedings/snp2021](#)]
7. **Yoshini Bailung**, on behalf of ALICE Collaboration; “*Measurements of heavy-flavor production as a function of multiplicity with ALICE at the LHC*”; **SciPost Physics Proceedings of the 50th International Symposium on Multiparticle Dynamics (ISMD2021), 12-16 July, 2021** [[10.21468/SciPostPhysProc.10.033](#)]
8. **Yoshini Bailung**, on behalf of ALICE Collaboration; “*Measurement of D-meson production as a function of charged-particle multiplicity in proton-proton collisions at $\sqrt{s} = 13$ TeV with ALICE at the LHC*”; Published in **Proceedings of the 9th Annual Large Hadron Collider Physics (LHCP2021) conference, 7 - 12 June, 2021** [[pos.sissa.it/397/190](#)]

TALKS

1. “*Searching for collective-like effects for heavy-flavour in small systems with ALICE*” at the **14 International Workshop on Multiple Parton Interactions at the LHC, MPI@LHC 2023**, University of Manchester, Manchester, United Kingdom, 20-24 November 2023; [[Link to contribution](#)] [Parallel]
2. “*Measurement of heavy-flavour production as a function of charge-particle multiplicity with ALICE at the LHC*” at the **4th Heavy Flavour Meet 2023**, IIT Goa, Goa, India,

2-4 November 2023; [\[Link to contribution\]](#) [Parallel]

3. “*Measurement of heavy-flavor production as a function of charge-particle multiplicity with ALICE at the LHC*” at the **9th edition of the Workshop for Young Scientists on the Physics of Ultra-relativistic Nucleus-Nucleus Collisions (Hot Quarks 2022)**, Colorado, USA, 11-17 October 2022; [\[Link to contribution\]](#) [Parallel]
4. “*Measurements of heavy-flavor production as a function of multiplicity with ALICE at the LHC*” at the **50th International Symposium on Multiparticle Dynamics (ISMD2021)**, virtual, 12-16 July, 2021; [\[Link to contribution\]](#) [Parallel]

POSTERS

1. Rohit Kaundal, **Yoshini Bailung**, Swapnesh Khade, Ankhi Roy;
“*A new look into ep collisions using event shape variables*” at the **68th DAE Symposium on Nuclear Physics, 7 - 11 December 2024, IIT Roorkee, Uttarakhand, India**
2. Swapnesh Khade, **Yoshini Bailung**, Rashi Gupta, Ravindra Singh, Ankhi Roy;
“*Forward Physics of D^0 meson: A study of production and angular correlations in pp and p -Pb collisions*” at the **68th DAE Symposium on Nuclear Physics, 7 - 11 December 2024, IIT Roorkee, Uttarakhand, India**
3. **Yoshini Bailung**, Sudhir Pandurang Rode, Neha Shah, Ankhi Roy;
“*Production of light nuclei in heavy-ion collisions via a coalescence mechanism*” at the **67th DAE Symposium on Nuclear Physics, 9 - 13 December 2023, IIT Indore, Indore, India**
4. Swapnesh Khade, Ravindra Singh, **Yoshini Bailung**, Ankhi Roy;
“*Influence of parton distribution functions on $D^0 - \bar{D}^0$ azimuthal angular correlations*” at the **4th Heavy Flavour Meet, 2 - 4 November 2023, IIT Goa, Goa, India**
5. Sumit K. Kundu, **Yoshini Bailung**, Ravindra Singh, Sudhir P. Rode, Ankhi Roy;
“*Deuteron production using UrQMD model via a coalescence afterburner at SPS energies*” at the **ICPAQGP, 7 - 10 February 2023, Puri, Odisha, India**
6. **Yoshini Bailung**, on behalf of ALICE Collaboration;
“*Measurement of D-meson production as a function of charged-particle multiplicity in proton-proton collisions at $\sqrt{s} = 13$ TeV with ALICE at the LHC*” at the **29th International Conference on Ultra-relativistic Nucleus-Nucleus Collisions (Quark Matter)**, 4 - 10 April, 2022, Krakow, Poland; [\[Link to contribution\]](#)
7. **Yoshini Bailung**, on behalf of ALICE Collaboration;
“*Measurement of D-meson production as a function of charged-particle multiplicity in proton-proton collisions at $\sqrt{s} = 13$ TeV with ALICE at the LHC*” at the **9th Annual Large Hadron Collider Physics (LHCP2021) conference, 7 - 12 June, 2021; [\[Link to contribution\]](#)**

RESEARCH SOFTWARE AND OPEN-SOURCE OUTPUTS

1. BDT b-jet tagging analysis tools [\[Link\]](#)
Python/ML workflow for BDT training, validation, template construction, flavor template fitting for post processing, template fitting
2. COALA : COALescence Afterburner [\[Link\]](#)
Event generator independent afterburner with phase-space and Wigner coalescence for deuteron, triton and helium-3 production.

MENTORING

1. Clement Lotteau [PhD student 2025 - Ongoing], IP2I Lyon, UCBL 1
Thesis title: *"Measurement of beauty production in proton-proton and Pb-Pb or OO collisions with ALICE at the LHC"*
2. Maelle Mazouni [License 3 stage (Internship), Duration: 6 weeks], Université Jean-Monnet
Project title: *"Physique des collisions d'ions lourds et phénoménologie du Plasma de Quarks et de Gluons"*
3. Priyanshu Khemka [Masters Student 2024 - 2025], IIT Indore
Thesis Title: *"Investigation of Gluon Saturation using Di-Hadron Correlation and assessing the efficiency and resolution of the EPUC detector in EIC"*
4. Rohit Kaundal [Masters Student 2023 - 2024], IIT Indore
Thesis Title: *"Investigating two-particle correlations in ep photoproduction and assessing the resolution and efficiency of the ePIC detector"*
5. Shankar Nair [Masters Student 2022 - 2023], IIT Indore
Thesis Title : *"Simulation studies of π^0 , η and ω meson reconstruction performance in pp collisions at $\sqrt{s} = 14$ TeV in Forward Calorimeter (FoCal), ALICE"*
6. Anurag [Masters Student 2022 - 2023], IIT Indore
Thesis Title: *"Flavour-dependent study of effects of multiple parton interactions and colour reconnection on the jets"*
7. Diksha Sharma [Masters Student 2021 - 2022], IIT Indore
Thesis Title : *"Feasibility Study of Exotic Particle $X(3872)$ with ATHENA in EIC Experiment and its Possible Structures"*

TEACHING

1. Teaching Assistant for Undergraduate and Graduate courses at IIT Indore
 - (a) Undergraduate Physics Lab [Autumn 2019, Autumn 2022]
 - (b) Electrodynamics [Autumn 2020]
 - (c) Numerical Methods in Physics [Autumn 2021]
 - (d) Mathematical Physics [Spring 2020]
2. Instructor in Machine Learning Using Python workshop organized by MNIT Jaipur

TECHNICAL AND PERSONAL SKILLS

Operating Systems: Linux, macOS, WINDOWS

Programming Languages: C/C++, python, FORTRAN, Bash, JavaScript, GDScript

Scientific Packages/Frameworks: ROOT, Mathematica

Specialised Packages/Frameworks: AliPhysics, PYTHIA, AMPT, UrQMD, SMASH

TypeSetting/ Web Development: LaTeX, HTML, CSS, Hugo

Spoken Languages: English, Hindi, Assamese (Mother tongue)